

A COMPLEMENTED-SUBSPACE OBSTRUCTION FOR UNIVERSAL NORM ATTAINMENT

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ABSTRACT. We record a structural consequence of the strictly singular ideal obstruction to Ostrovskii's problem. If an infinite-dimensional Banach space X contains a complemented infinite-dimensional subspace with the bounded compact approximation property, then some operator on X fails to attain its norm. Equivalently, any infinite-dimensional space on which every operator attains its norm must have the following strong projection obstruction: every complemented infinite-dimensional subspace fails the bounded compact approximation property. This does not solve Ostrovskii's problem, but it pushes the obstruction below the level of global approximation properties of X itself.

1. SOURCE PROBLEM

The source paper is:

Sheldon Dantas, Mingu Jung, Gonzalo Martinez-Cervantes, *On the existence of non-norm-attaining operators*, arXiv:2102.06452.

The paper quotes the following problem of Ostrovskii.
from Lemma 3.4. □

M.I. Ostrovskii asked in [23, §12, pg. 65] whether there exist infinite dimensional Banach spaces on which every operator attains its norm (this question is also asked in [19, Problem 8] and [14, Problem 217]). By Holub's Theorem, if such an infinite dimensional Banach space exists, it cannot have the AP. Theorem 3.8 below should be seen as a generalization of this fact. Let us recall that given a (norm-closed) operator ideal $\mathcal{A}$ and $\lambda \geq 1$, a Banach space E is said to have the λ - $\mathcal{A}$ -approximation property (for short, λ - $\mathcal{A}$ -AP) if the identity operator Id_E belongs to $\overline{\{T \in \mathcal{A}(E, E) : \|T\| \leq \lambda\}}^{\tau_c}$. We say that E has the bounded- $\mathcal{A}$ -AP if it has the λ - $\mathcal{A}$ -AP for some $\lambda \geq 1$. This general approximation property has been studied, for instance, in [12, 20, 25, 28].

Theorem 3.8. *If there is an infinite dimensional Banach space E such that every operator on $\mathcal{L}(E)$ attains its norm, then E does not have the bounded $\mathcal{A}$ -approximation property for any ideal $\mathcal{A}$ not containing the identity on E .*

In words, the problem asks whether there exists an infinite-dimensional Banach space X such that every bounded operator $X \rightarrow X$ attains its norm. Dantas–Jung–Martinez-Cervantes prove that such a space, if it exists, cannot have the bounded $\mathcal{A}$ -approximation property for any norm-closed operator ideal $\mathcal{A}$ not containing the identity [1, Theorem 3.8].

The observation below gives a different kind of obstruction. It does not ask that X itself have a bounded approximation property. It is enough that X contain a complemented infinite-dimensional subspace with bounded compact approximation.

2. DEFINITIONS

Definition 1. Let $\mathcal{S}(E, F)$ denote the class of strictly singular operators from E to F . Thus $T \in \mathcal{S}(E, F)$ if no restriction of T to an infinite-dimensional subspace of E is an isomorphism onto its range.

Definition 2. A Banach space Y has the bounded compact approximation property, abbreviated BCAP, if there is $\lambda \geq 1$ such that

$$\text{Id}_Y \in \overline{\{K \in \mathcal{K}(Y) : \|K\| \leq \lambda\}}^{\tau_c},$$

where τ_c is the topology of compact convergence.

3. PROOF INTUITION

The strictly singular ideal gives a useful way to test whether the unit ball of an operator ideal is closed in the strong operator topology. If X had universal norm attainment, the Dantas–Jung–Martinez–Cervantes James-property criterion forces the strictly singular unit ball in $\mathcal{L}(X)$ to be SOT closed.

Now suppose $P : X \rightarrow Y \subset X$ is a bounded projection onto an infinite-dimensional subspace Y with BCAP. Compact approximants to Id_Y , transported through P and the inclusion $Y \hookrightarrow X$, give compact, hence strictly singular, operators on X . These operators are uniformly bounded and converge strongly to P . But P is not strictly singular because it is the identity on the infinite-dimensional range Y . Thus the strictly singular ball is not SOT closed, contradicting universal norm attainment.

4. MAIN RESULT

Lemma 1. Let X be reflexive. If every operator in $\mathcal{L}(X)$ attains its norm, then the unit ball of $\mathcal{S}(X)$ is SOT closed in $\mathcal{L}(X)$.

Proof. We spell out the standard reduction because this is the point where strict singularity enters. The ideal of strictly singular operators is norm closed and is separably determined in the domain variable: if T is not strictly singular, then T is bounded below on an infinite-dimensional subspace, and any separable infinite-dimensional subspace of that subspace still witnesses non-strict singularity.

If $B_{\mathcal{S}(X)}$ were not SOT closed, the separable-complementation argument used in [1, Lemma 3.5] would turn this SOT non-closure into SOT sequential non-closure. Indeed, reflexivity gives the separable complementation property, and strict singularity is detected on separable subspaces as above.

The set $B_{\mathcal{S}(X)}$ is norm closed and convex. Therefore [1, Lemma 3.4] and [1, Theorem A] imply that SOT sequential non-closure of this ball yields a non-norm-attaining operator on X , a contradiction. Hence $B_{\mathcal{S}(X)}$ is SOT closed. $\square$

Theorem 1. Let X be an infinite-dimensional Banach space. Suppose that X contains a complemented infinite-dimensional subspace Y with the bounded compact approximation property. Then there exists an operator on X which does not attain its norm.

Proof. If X is not reflexive, James' theorem gives a non-norm-attaining functional on X , and hence a non-norm-attaining rank-one operator on X . Thus assume X is reflexive.

Let $P : X \rightarrow Y$ be a bounded projection and let $J : Y \hookrightarrow X$ be the inclusion. Since Y has BCAP, choose $\lambda \geq 1$ and a net $(K_\alpha) \subset \mathcal{K}(Y)$ such that

$$\|K_\alpha\| \leq \lambda \quad \text{and} \quad K_\alpha y \rightarrow y \quad (y \in Y).$$

For each α , define

$$S_\alpha = JK_\alpha P \in \mathcal{L}(X).$$

Each S_α is compact, hence strictly singular, and $\|S_\alpha\| \leq \lambda\|P\|$. Also $S_\alpha x \rightarrow J \text{Id}_Y Px = Px$ for every $x \in X$. Hence

$$(\lambda\|P\|)^{-1}S_\alpha \longrightarrow (\lambda\|P\|)^{-1}P \quad \text{in SOT.}$$

The operator P is not strictly singular, since $P|_Y = \text{Id}_Y$ and Y is infinite dimensional. Therefore the unit ball of $\mathcal{S}(X)$ is not SOT closed. By Lemma 1, it is impossible that every operator on X attain its norm. Thus some operator on X is non-norm-attaining. $\square$

Remark 1. *Every Banach space with the bounded approximation property has BCAP. Therefore Theorem 1 applies, in particular, when X contains a complemented infinite-dimensional subspace with a Schauder basis.*

5. VERIFICATION AND SCOPE

The proof uses three standard facts:

- compact operators are strictly singular;
- strict singularity is a norm-closed, domain-separably-determined operator ideal;
- the Dantas–Jung–Martinez–Cervantes James-property criterion forces SOT closedness of the strictly singular unit ball under universal norm attainment.

This result does not settle Ostrovskii’s problem. It says that a positive example, if one exists, must be highly resistant to complemented approximation-property structure: no complemented infinite-dimensional subspace may have BCAP. In particular, the usual route through a complemented basic sequence is impossible for a candidate.

6. NOVELTY CHECK

The cheap run indexes were searched for arXiv:2102.06452, Ostrovskii, strictly singular, bounded compact approximation, and complemented subspace phrases. The only direct run hit was the existing partial packet on separably determined ideal obstructions. Web searches for combinations of “complemented subspace”, “bounded compact approximation property”, and “non-norm-attaining operator” did not find this exact corollary.

The novelty should still be regarded as modest. The source paper’s arXiv TeX comments already point out that the strictly singular ideal can be inserted into the SOT-closure argument. The contribution here is the explicit complemented-subspace consequence and its use as an attack note for Ostrovskii’s problem.

7. HUMAN REVIEW RECOMMENDATION

Review as a likely valid partial obstruction. The main points to check are:

- the normalization of the compact approximants $JK_\alpha P$;
- that P is indeed outside the strictly singular ideal;
- that the invocation of the Dantas–Jung–Martinez–Cervantes SOT-closure criterion for the strictly singular ball is legitimate.

REFERENCES

- [1] S. Dantas, M. Jung, and G. Martinez-Cervantes, *On the existence of non-norm-attaining operators*, arXiv:2102.06452.
- [2] G. Martinez-Cervantes, M. Jung, and A. Rueda Zoca, *Rank-one perturbations and norm-attaining operators*, arXiv:2301.05003.